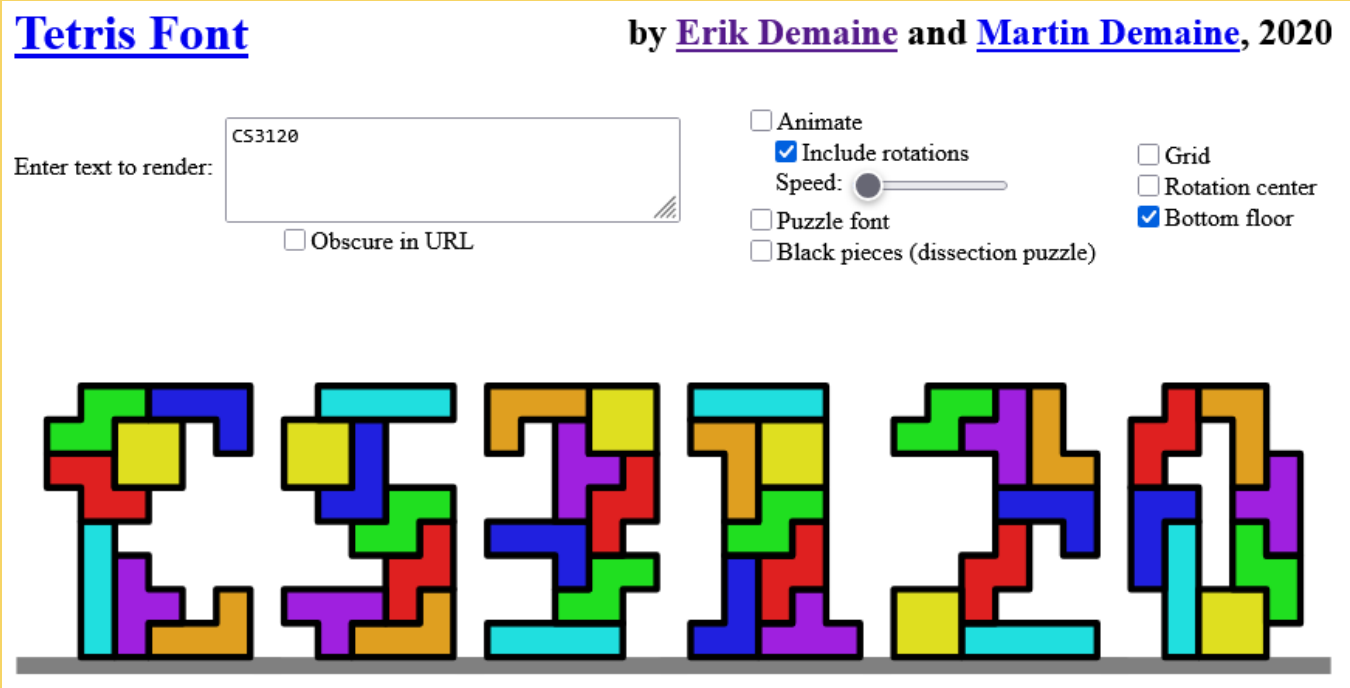
HW 9 due next Monday.
Quiz 12 due next Monday.



https://erikdemaine.org/fonts/tetris/?text=CS3120

Class 27:
P vs NP, Implications

University of Virginia
CS3120: DMT2
https://weikailin.github.io/cs3120-toc
Wei-Kai Lin

Plan

NP Problems, Implication

Cryptography

NP-Complete Problems

Independent Set

Graph 3-Coloring

Textbook [TCS] Section 15.2.1, 14.5

https://introtcs.org/public/lec_13_Cook_Levin.html
NP, NP Completeness

Gemini generated cover image.

Recap

Cook-Levin Theorem

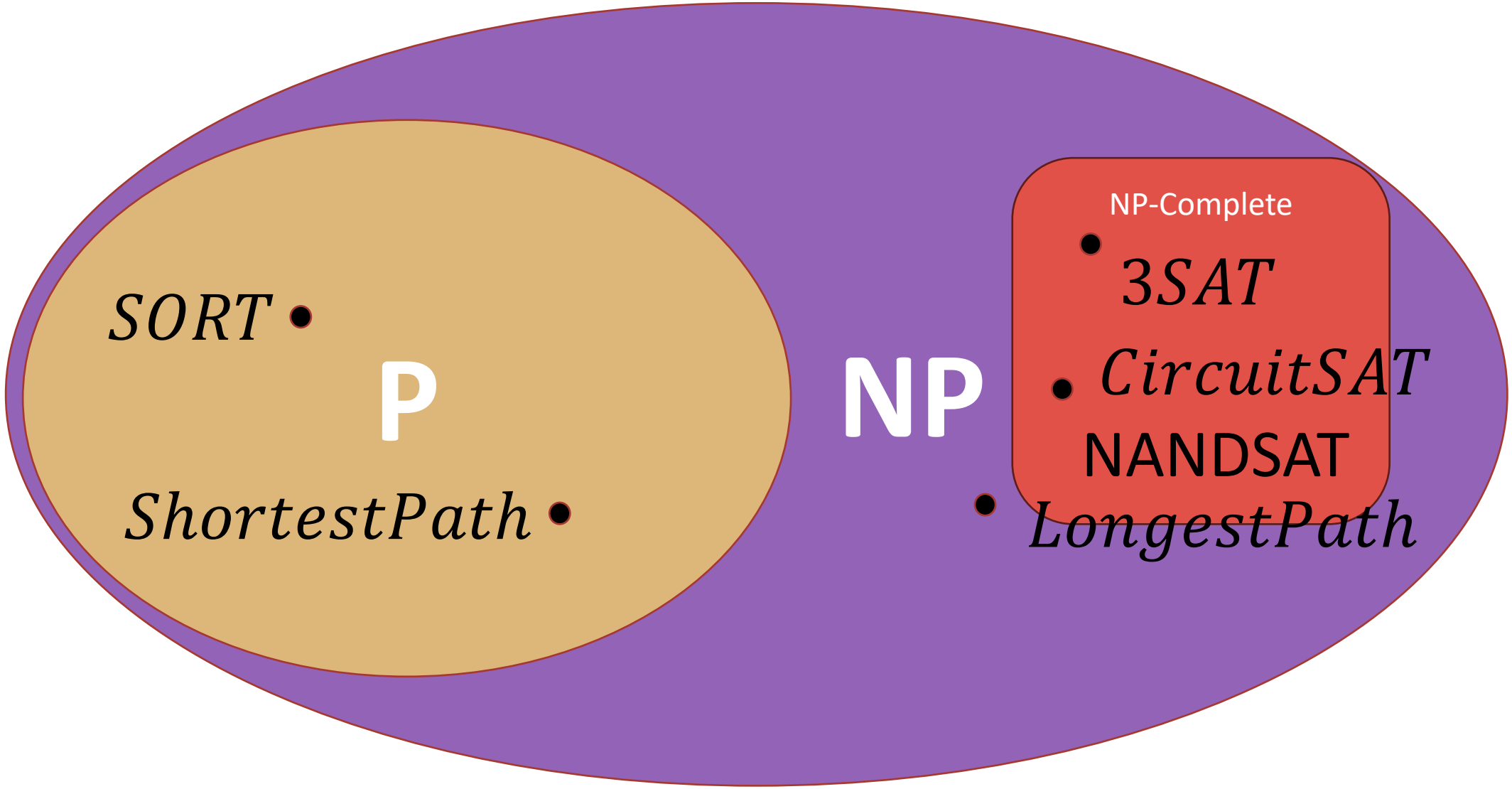
(Theorem 15.6 in the TCS Book):

For every $F \in \mathbf{NP}$, $F \leq_p$ **3SAT.**

Definition: A Boolean function G is **NP-Hard** if every $F \in \mathbf{NP}$ can be reduced to G: $F \leq_p G$.

Definition: A function G is **NP-Complete** if $G \in \mathbf{NP}$ and G is **NP-Hard**.

A function $F: \{0,1\}^* \to \{0,1\}$ is in **NP** if there exists some $a \in \mathbb{N}^+$ and $V: \{0,1\}^* \to \{0,1\}$ such that $V \in \mathbf{P}$ and $\forall x \in \{0,1\}^n$, $F(x) = 1 \Leftrightarrow \exists w \in \{0,1\}^{n^a}$ such that $V(x,w) = 1$.

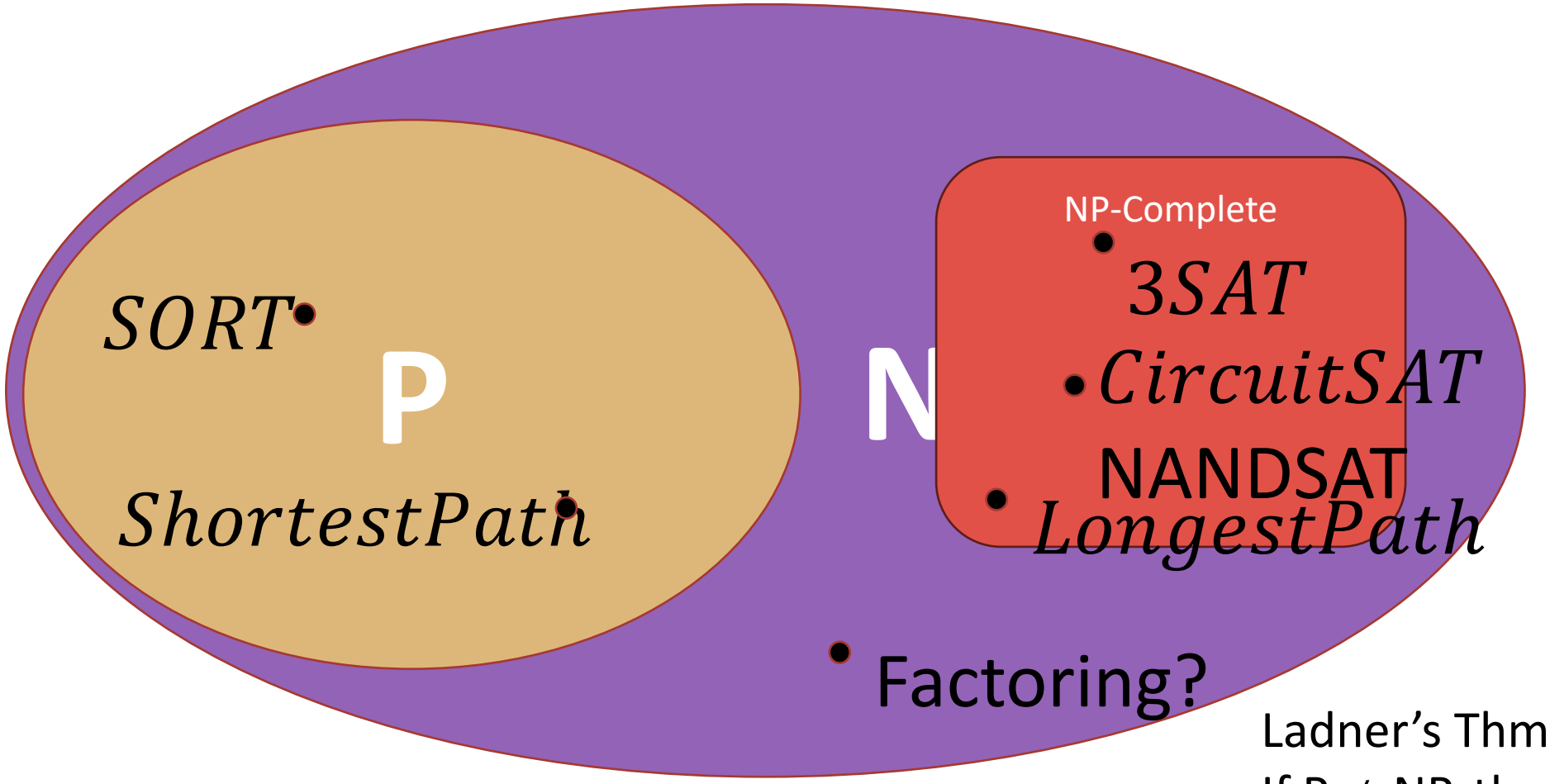


Unknown if $3SAT \in$ **P**

Known that $3SAT \in$ **NP**

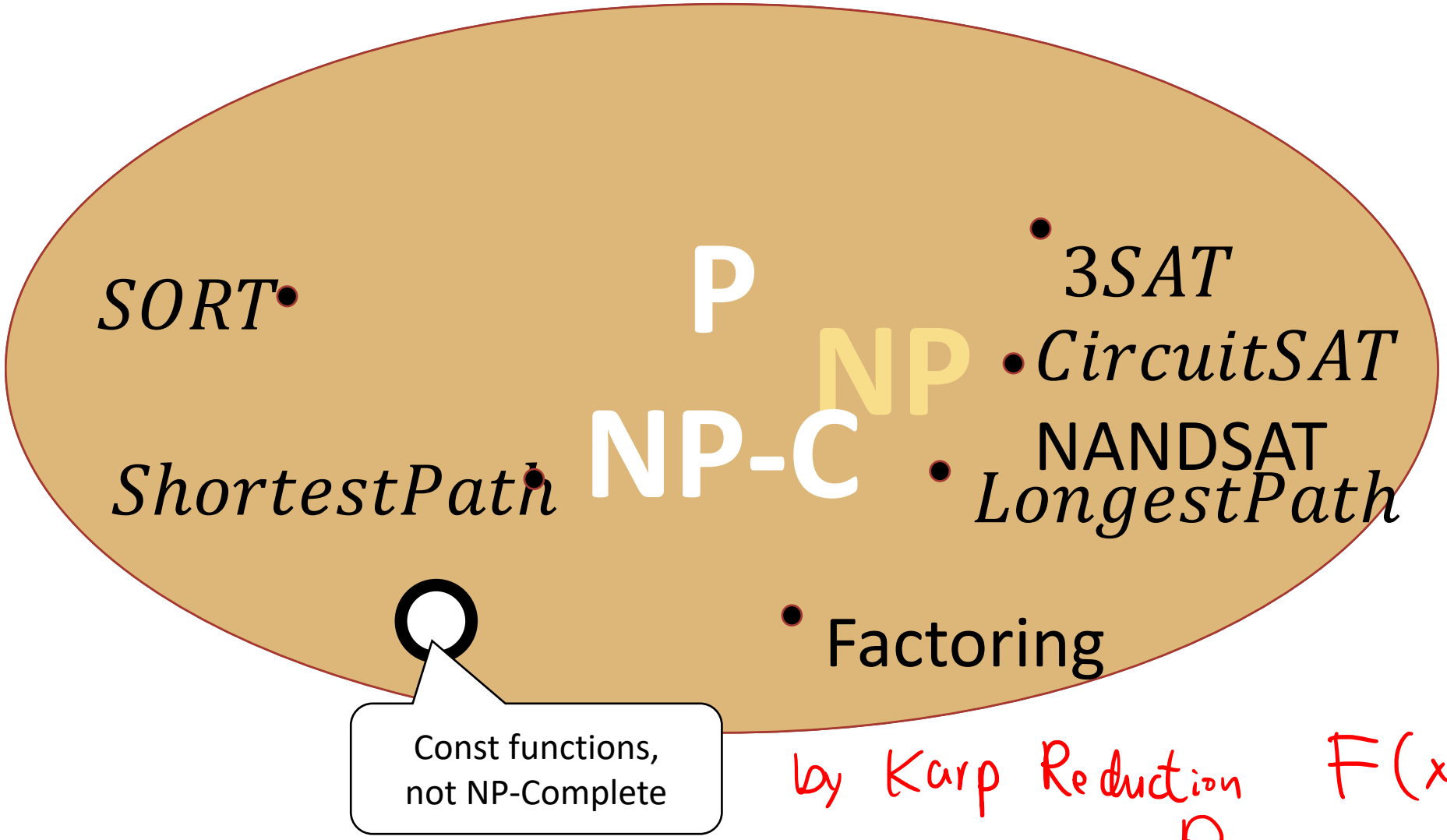
Consequences

P ≠ NP



Ladner's Thm:
If P ≠ NP, there exists a NP problem
not in **P** and not **NP**-Complete

P = NP

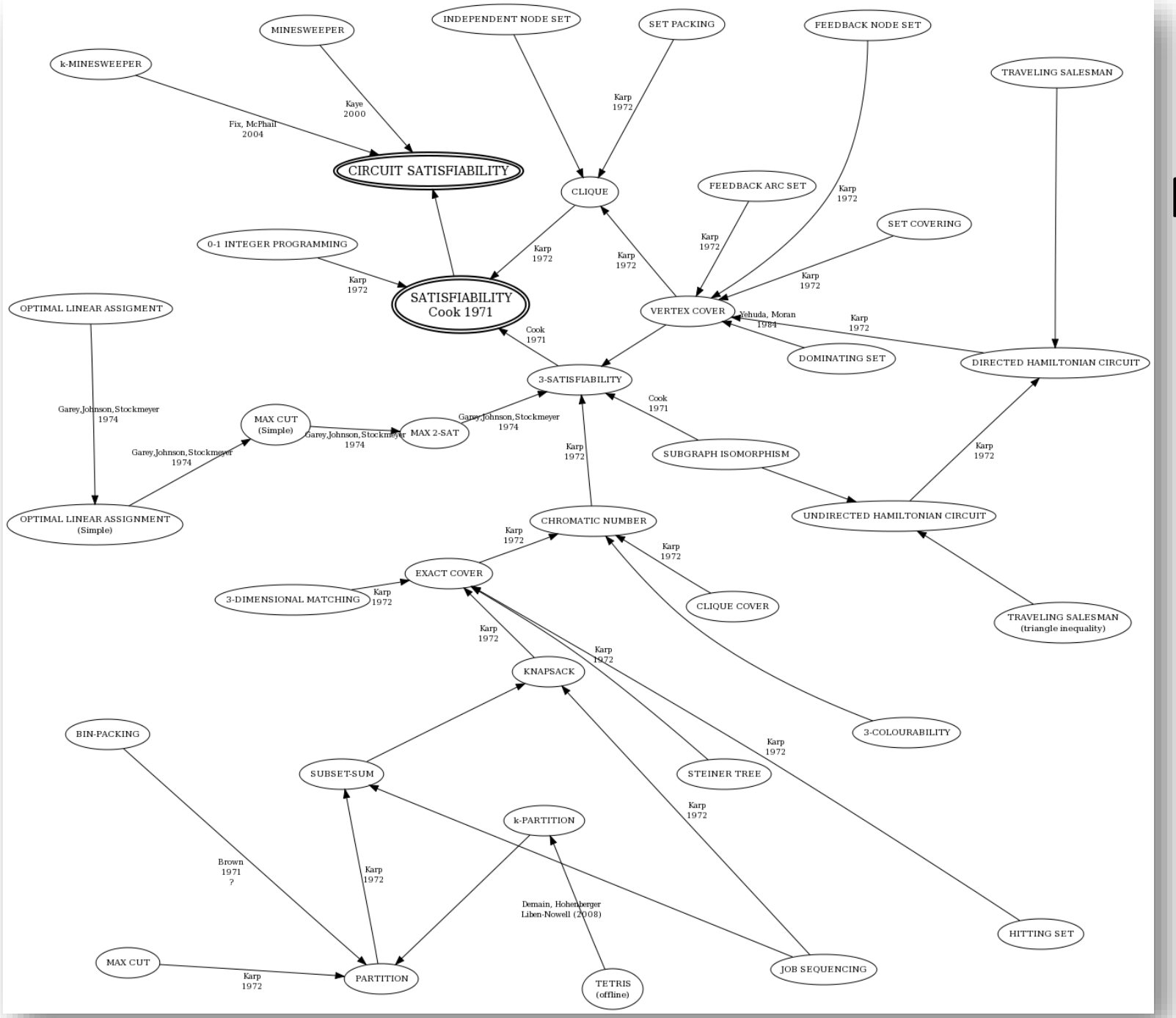


by Karp Reduction R

$F(x) = G(R(x))$

More NP-Complete problems

NP-Complete Problems



https://donsbot.com/2011/01/16/painless-np-complete-problems-an-embedded-dsl-for-smt-solving/

Some NP-Complete Problems

BinPacking

Input: Finite set of items, I, size $s(i)$ for each item $i \in I$, bin capacity B, number of bins k

Output: 1 iff there is a partition $I_1, \ldots, I_k$ such that the total size of I_j is at most B

Traveling Salesman Problem

Input: A weighted graph $G = (V, E), w(e \in E) \rightarrow \mathbb{N}$, a start node $s \in V$, a cost z

Output: 1 iff there path in G that visits every node with cost $\leq z$

SetCover

Input: A set $U, S \subseteq Pow(U)$ where $\bigcup_{s \in S} s = U$, $m \in \mathbb{N}$

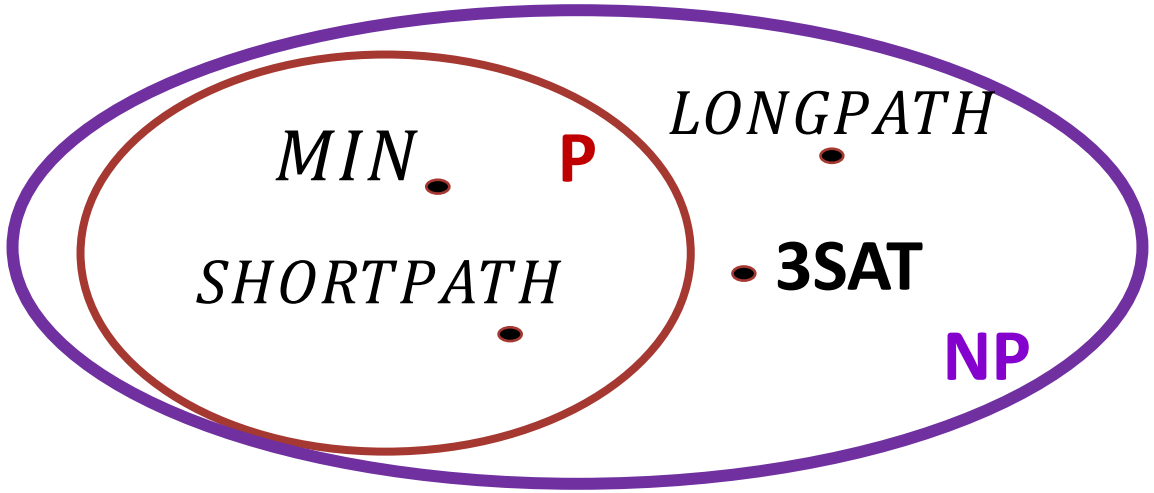
Output: 1 iff there is a subset S' of S where $\bigcup_{s \in S'} s = U$ and $|S'| \leq m$.

Which outcome do we prefer?

P = NP

3SAT ∈ P

P ⊊ NP



3SAT ∉ P

Do we "solve" these okay today?

BinPacking

Input: Finite set of items, I, size $s(i)$ for each item $i \in I$, bin capacity B, number of bins k

Output: 1 iff there is a partition $I_1, \ldots, I_k$ such that the total size of I_j is at most B

Traveling Salesman Problem

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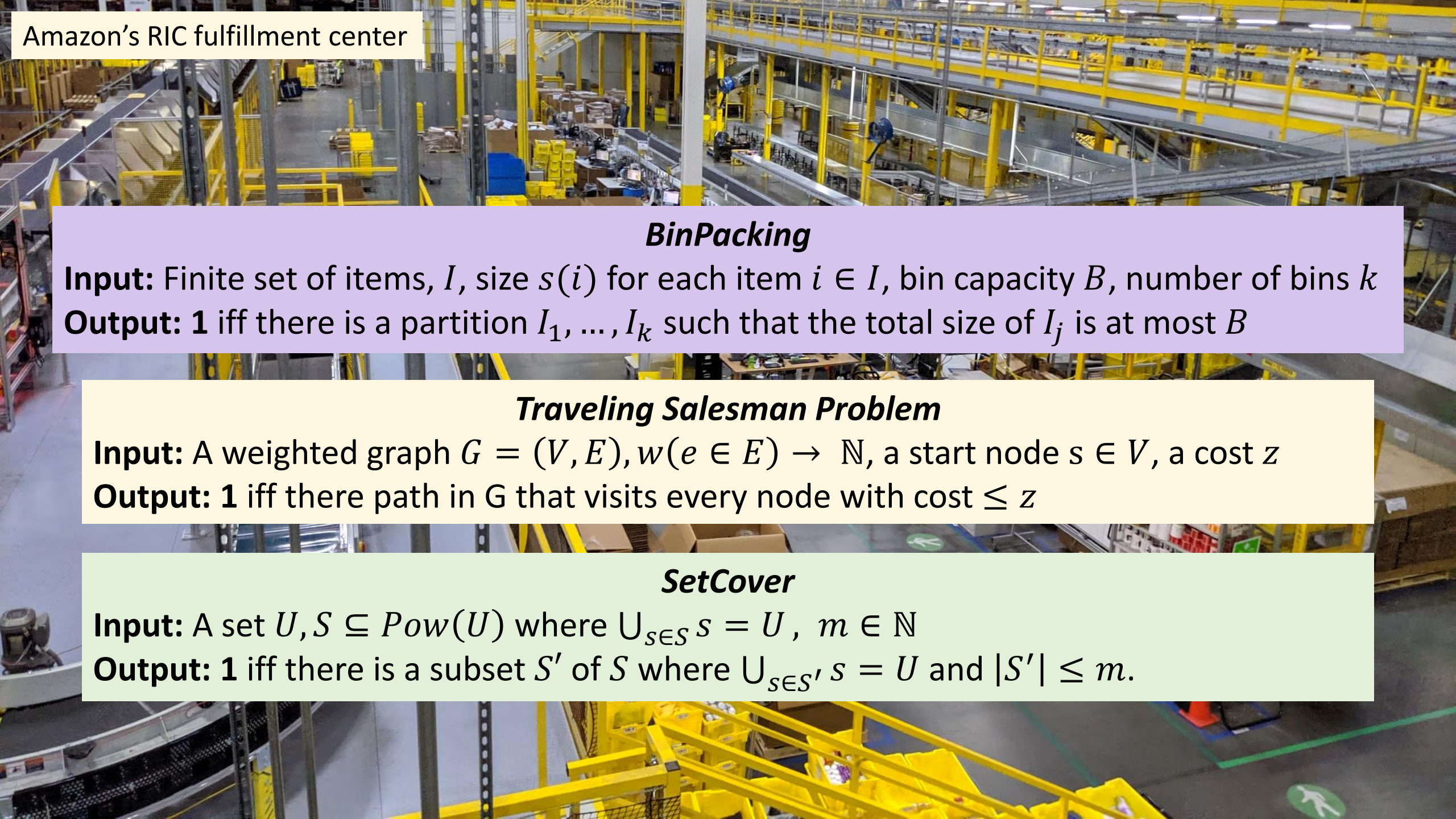
Amazon’s RIC fulfillment center



Amazon's RIC fulfillment center

BinPacking

Input: Finite set of items, I, size $s(i)$ for each item $i \in I$, bin capacity B, number of bins k

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Traveling Salesman Problem

Input: A weighted graph $G = (V, E), w(e \in E) \rightarrow \mathbb{N}$, a start node $s \in V$, a cost z

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SetCover

Input: A set $U, S \subseteq Pow(U)$ where $\bigcup_{s \in S} s = U$, $m \in \mathbb{N}$

Output: 1 iff there is a subset S' of S where $\bigcup_{s \in S'} s = U$ and $|S'| \leq m$.

What about 3SAT?

50 variables, 100 random clauses

$$(x_{48} \vee x_4 \vee \overline{x_9}) \wedge (\overline{x_{44}} \vee x_{50} \vee \overline{x_{37}}) \wedge (\overline{x_8} \vee \overline{x_1} \vee x_{28}) \wedge (x_{21} \vee x_{27} \vee \overline{x_{32}}) \wedge (x_{17} \vee x_{29} \vee \overline{x_{30}}) \wedge (x_{30} \vee$$
$$x_{24} \vee x_{25}) \wedge (\overline{x_{22}} \vee \overline{x_{27}} \vee \overline{x_{45}}) \wedge (x_{25} \vee \overline{x_{24}}) \wedge (\overline{x_{44}} \vee x_{50} \vee x_{14}) \wedge (x_{45} \vee x_{15} \vee x_{37}) \wedge (\overline{x_{16}} \vee x_{14} \vee$$
$$\overline{x_{36}}) \wedge (x_{33} \vee x_5 \vee x_{26}) \wedge (x_{18} \vee x_7 \vee \overline{x_{24}}) \wedge (x_{31} \vee x_{38} \vee x_{28}) \wedge (x_{31} \vee \overline{x_{33}} \vee \overline{x_8}) \wedge (x_{49} \vee x_7 \vee \overline{x_6}) \wedge$$
$$(x_{34} \vee \overline{x_8} \vee x_{46}) \wedge (x_4 \vee \overline{x_5} \vee \overline{x_{35}}) \wedge (x_{43} \vee x_{27} \vee x_{39}) \wedge (\overline{x_{46}} \vee \overline{x_{40}} \vee \overline{x_{27}}) \wedge (\overline{x_{25}} \vee x_{14} \vee \overline{x_{49}}) \wedge (x_{38} \vee$$
$$x_5 \vee x_{15}) \wedge (x_9 \vee x_{14} \vee \overline{x_{19}}) \wedge (x_{45} \vee \overline{x_{42}} \vee \overline{x_{39}}) \wedge (x_{34} \vee \overline{x_{22}} \vee \overline{x_{28}}) \wedge (\overline{x_{20}} \vee x_{15} \vee \overline{x_8}) \wedge (\overline{x_{44}} \vee \overline{x_{10}} \vee$$
$$\overline{x_9}) \wedge (x_{22} \vee \overline{x_{31}} \vee x_{14}) \wedge (\overline{x_9} \vee \overline{x_{42}} \vee \overline{x_{15}}) \wedge (\overline{x_{40}} \vee x_{12} \vee \overline{x_{32}}) \wedge (\overline{x_{20}} \vee \overline{x_6} \vee \overline{x_{15}}) \wedge (\overline{x_{37}} \vee x_{39} \vee \overline{x_{23}}) \wedge$$
$$(\overline{x_3} \vee \overline{x_{40}} \vee \overline{x_{32}}) \wedge (\overline{x_4} \vee \overline{x_{25}} \vee x_7) \wedge (\overline{x_{20}} \vee \overline{x_{36}} \vee \overline{x_{37}}) \wedge (\overline{x_{40}} \vee \overline{x_{35}} \vee x_{39}) \wedge (\overline{x_{43}} \vee \overline{x_{40}} \vee \overline{x_7}) \wedge (x_{34} \vee$$
$$x_{44} \vee x_{26}) \wedge (x_{13} \vee x_{27} \vee x_{28}) \wedge (x_{12} \vee \overline{x_{36}} \vee x_7) \wedge (\overline{x_{16}} \vee x_9 \vee \overline{x_{24}}) \wedge (\overline{x_{48}} \vee x_{14} \vee x_{28}) \wedge (x_{16} \vee x_4 \vee$$
$$x_{40}) \wedge (\overline{x_{25}} \vee x_{15} \vee x_{26}) \wedge (\overline{x_{22}} \vee \overline{x_{46}} \vee x_{27}) \wedge (x_{20} \vee x_{49} \vee x_{11}) \wedge (x_{42} \vee \overline{x_{10}} \vee x_{28}) \wedge (\overline{x_{45}} \vee x_4 \vee x_{35} \vee \overline{x_{27}}) \wedge$$
$$(\overline{x_{22}} \vee x_{47} \vee x_{26}) \wedge (\overline{x_{22}} \vee \overline{x_{46}} \vee x_{27}) \wedge (x_{20} \vee x_{49} \vee x_{11}) \wedge (x_{42} \vee \overline{x_{10}} \vee x_{28}) \wedge (\overline{x_{45}} \vee x_{28} \vee \overline{x_{37}}) \wedge$$
$$(x_{14} \vee \overline{x_{32}} \vee \overline{x_{23}}) \wedge (x_{22} \vee x_{14} \vee x_{23}) \wedge (\overline{x_{17}} \vee \overline{x_{46}} \vee \overline{x_7}) \wedge (\overline{x_{31}} \vee x_{46} \vee \overline{x_{50}}) \wedge (x_{34} \vee \overline{x_{41}} \vee x_{43}) \wedge (x_{17} \vee$$
$$\overline{x_9} \vee x_{15}) \wedge (x_{46} \vee x_{14} \vee \overline{x_{12}}) \wedge (\overline{x_{20}} \vee x_{12} \vee x_{14}) \wedge (x_{41} \vee x_{42} \vee \overline{x_{15}}) \wedge (x_{48} \vee x_{46} \vee \overline{x_{36}}) \wedge (\overline{x_{22}} \vee \overline{x_4} \vee$$
$$\overline{x_{49}}) \wedge (x_{22} \vee x_{12} \vee \overline{x_{42}}) \wedge (x_{13} \vee \overline{x_{38}} \vee x_{39}) \wedge (x_{48} \vee \overline{x_{16}} \vee \overline{x_{27}}) \wedge (x_{17} \vee \overline{x_{18}} \vee \overline{x_{26}}) \wedge (x_{48} \vee \overline{x_{40}} \vee$$
$$\overline{x_{35}}) \wedge (\overline{x_{43}} \vee \overline{x_{40}} \vee \overline{x_{49}}) \wedge (x_{29} \vee x_{11} \vee \overline{x_{32}}) \wedge (x_{33} \vee \overline{x_{17}} \vee x_{39}) \wedge (\overline{x_{25}} \vee \overline{x_9} \vee \overline{x_6}) \wedge (x_{40} \vee \overline{x_{50}} \vee x_{19}) \wedge$$
$$(x_8 \vee x_{10} \vee \overline{x_{27}}) \wedge (x_5 \vee x_9 \vee \overline{x_{26}}) \wedge (x_{45} \vee \overline{x_{38}} \vee \overline{x_{27}}) \wedge (\overline{x_4} \vee \overline{x_{40}} \vee \overline{x_{42}}) \wedge (x_{21} \vee x_{50} \vee x_{12}) \wedge (\overline{x_8} \vee$$
$$\overline{x_{14}} \vee \overline{x_{42}}) \wedge (\overline{x_{17}} \vee x_{47} \vee \overline{x_{27}}) \wedge (x_{49} \vee \overline{x_{12}} \vee \overline{x_6}) \wedge (x_{27} \vee x_{49} \vee \overline{x_{32}}) \wedge (\overline{x_{29}} \vee \overline{x_{12}} \vee \overline{x_{26}}) \wedge (x_{48} \vee \overline{x_2} \vee$$
$$x_6) \wedge (x_{16} \vee x_{36} \vee x_{49}) \wedge (x_{33} \vee \overline{x_{12}} \vee \overline{x_{26}}) \wedge (\overline{x_{33}} \vee x_{29} \vee x_{49}) \wedge (\overline{x_{48}} \vee x_2 \vee x_{19}) \wedge (x_{25} \vee x_{36} \vee x_{49}) \wedge$$
$$(x_{21} \vee x_{40} \vee \overline{x_{14}}) \wedge (\overline{x_{34}} \vee \overline{x_{44}} \vee \overline{x_6}) \wedge (x_{48} \vee \overline{x_{50}} \vee \overline{x_1}) \wedge (x_5 \vee \overline{x_{12}} \vee x_7) \wedge (x_{21} \vee \overline{x_{35}} \vee \overline{x_{27}}) \wedge (\overline{x_{22}} \vee$$
$$\overline{x_{16}} \vee \overline{x_{14}}) \wedge (\overline{x_{13}} \vee \overline{x_{35}} \vee \overline{x_{12}}) \wedge (\overline{x_4} \vee \overline{x_{35}} \vee \overline{x_{42}}) \wedge (\overline{x_{50}} \vee \overline{x_{40}} \vee x_7) \wedge (x_{25} \vee x_{47} \vee \overline{x_{12}})$$

sahands / simple-sat

Watch 5 | Unstar 29 | Fork 5

Code | Issues 0 | Pull requests 0 | Wiki | Pulse | Graphs

Simple recursive and iterative SAT solver written in Python.

38 commits | 1 branch | 0 releases | 1 contributor

Branch: master | New pull request | Create new file | Upload files | Find file | Clone or download

sahands Moving article content. Latest commit b987fb0 on Sep 30, 2014

src	Added the post content to the repository.	2 years ago
.gitignore	Initial commit	2 years ago
LICENSE	Initial commit	2 years ago
README.rst	Renames	2 years ago

README.rst

sim

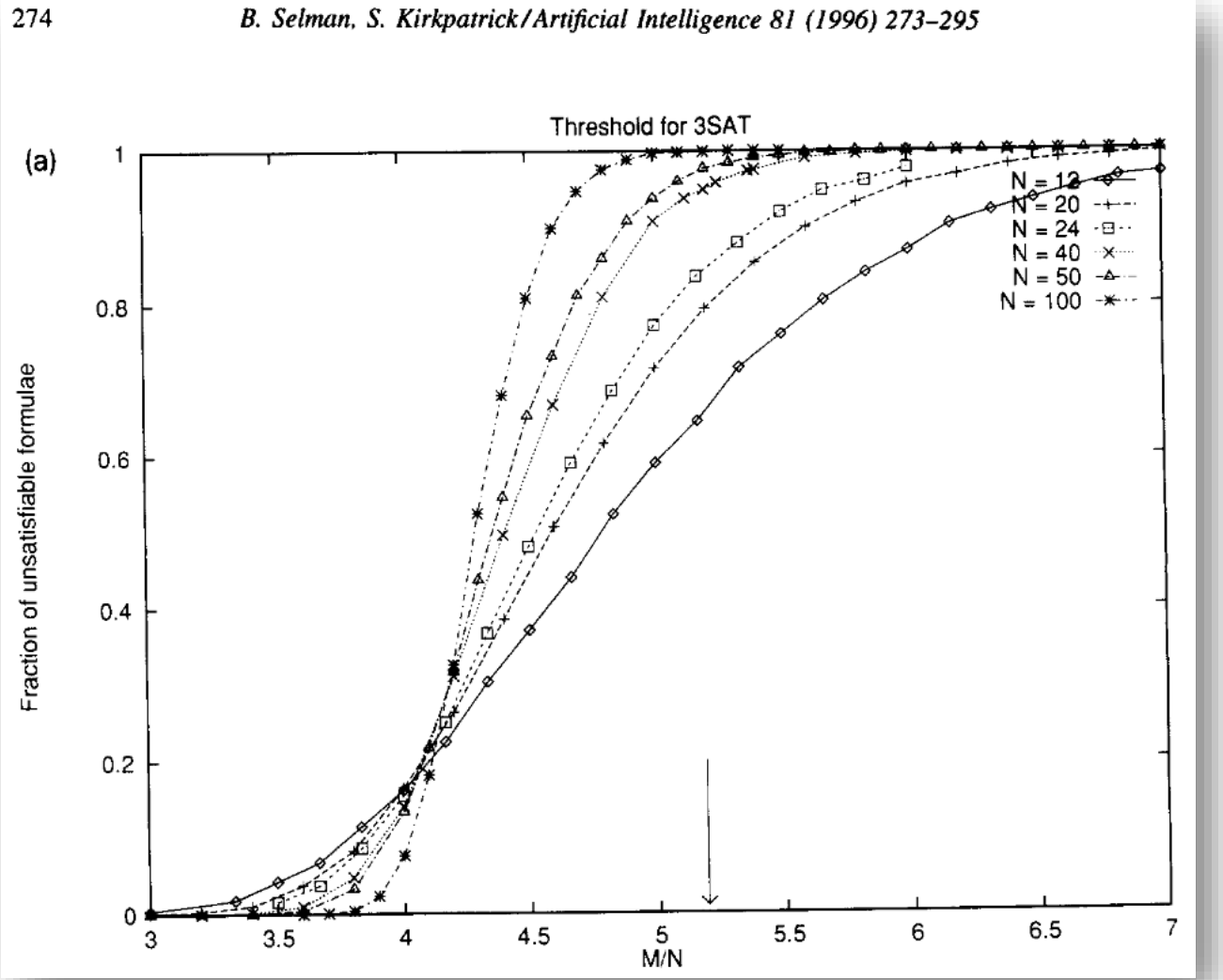
This p
based

```
> time python3 sat.py -i random_50_100.in
~x48 ~x4 ~x9 ~x37 ~x44 ~x50 ~x1 ~x28 ~x8 ~x21 ~x27 ~x32 ~x17 ~x29 ~x30 x24 ~x22
~x25 ~x14 ~x15 x45 ~x16 x36 ~x26 ~x33 ~x5 x18 x7 ~x31 x38 ~x49 ~x6 x34 x46 ~x35
x39 ~x43 x40 ~x19 ~x42 ~x20 ~x10 x12 x23 ~x3 x13 x47 ~x11 ~x41 ~x2

real    0m0.053s
user    0m0.042s
sys     0m0.009s
>
```

3SAT is “easy on average”!?

Well, it depends on the <u>distribution</u> and how <u>easy</u> we want.



N: num variables

M: num clauses

https://www.cs.cornell.edu/selman/papers/pdf/96.aij.critical_behavior_cost.pdf

If knowing **P=NP** *wouldn't help us much in solving practical problems we want to solve, would there be a benefit to knowing* **P ⊊ NP?**

Example: Encryption

Some details needed:
Key-length < message
Correct decryption

AES (Advanced Encryption Standard) is a poly-time function:
AES(key, message) ↦ ciphertext

Fix message = "secret Amazon order ….", define function:

$F(c) =$ 1 if exists k s.t. AES(k, message) = c
0 otherwise

witness

if P = NP, no encrypt

We have $F \in NP$ (why?)
If P = NP, what happened to message?

Hard Problems are Useful

P = NP would imply most cryptographic goals are impossible: no encryption / digital signature could exist!

Assumptions in Cryptography:

Factoring is **hard** $\Longrightarrow$ **RSA** is secure

"Lattice problems" are **hard** $\Longrightarrow$ some "post quantum" cryptosystems are secure

Inaccurate (ask me / cryptography course)

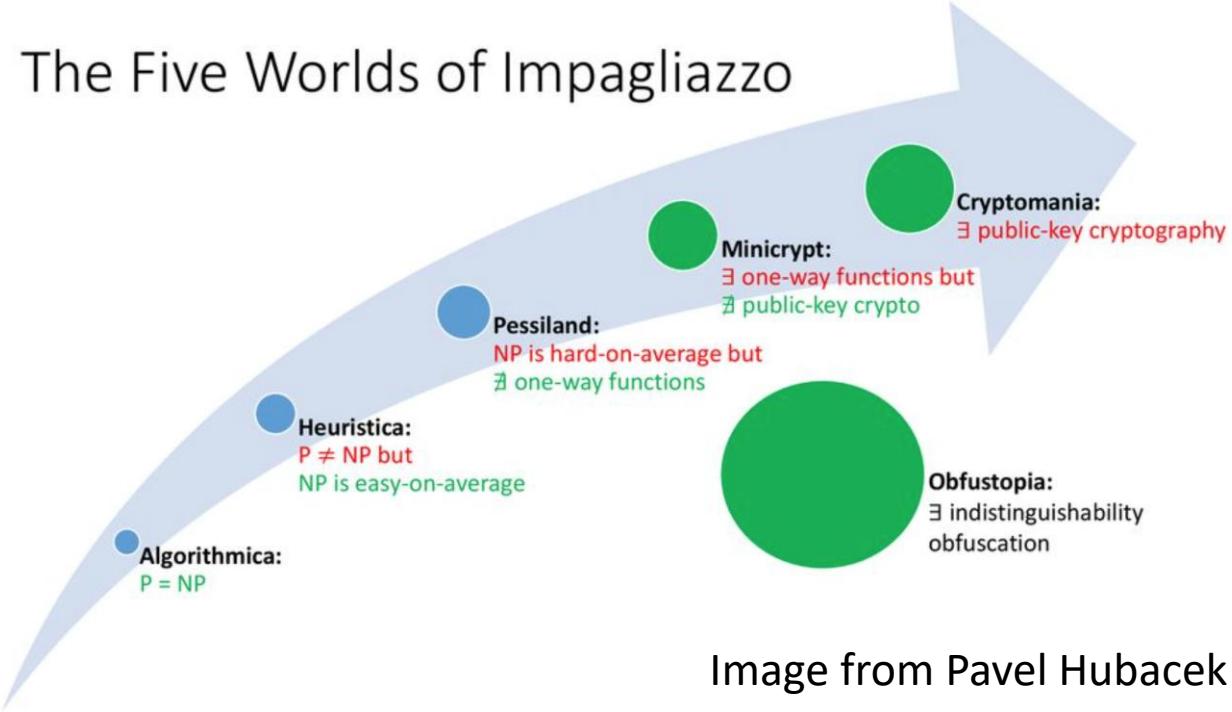
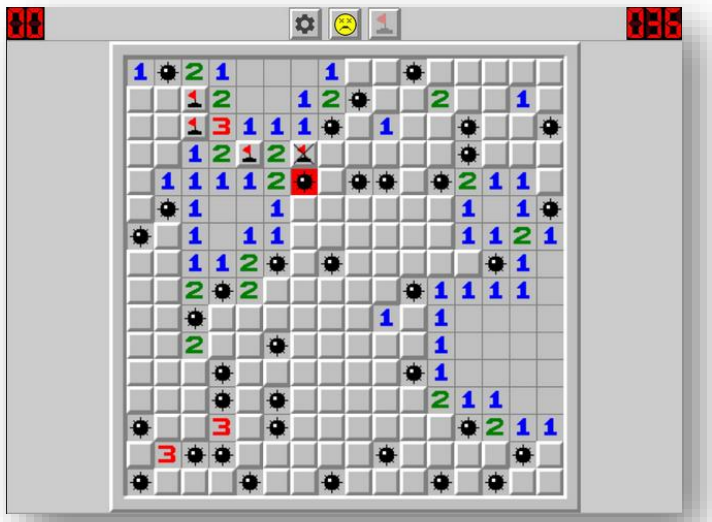


Image from Pavel Hubacek

NP-Complete Games



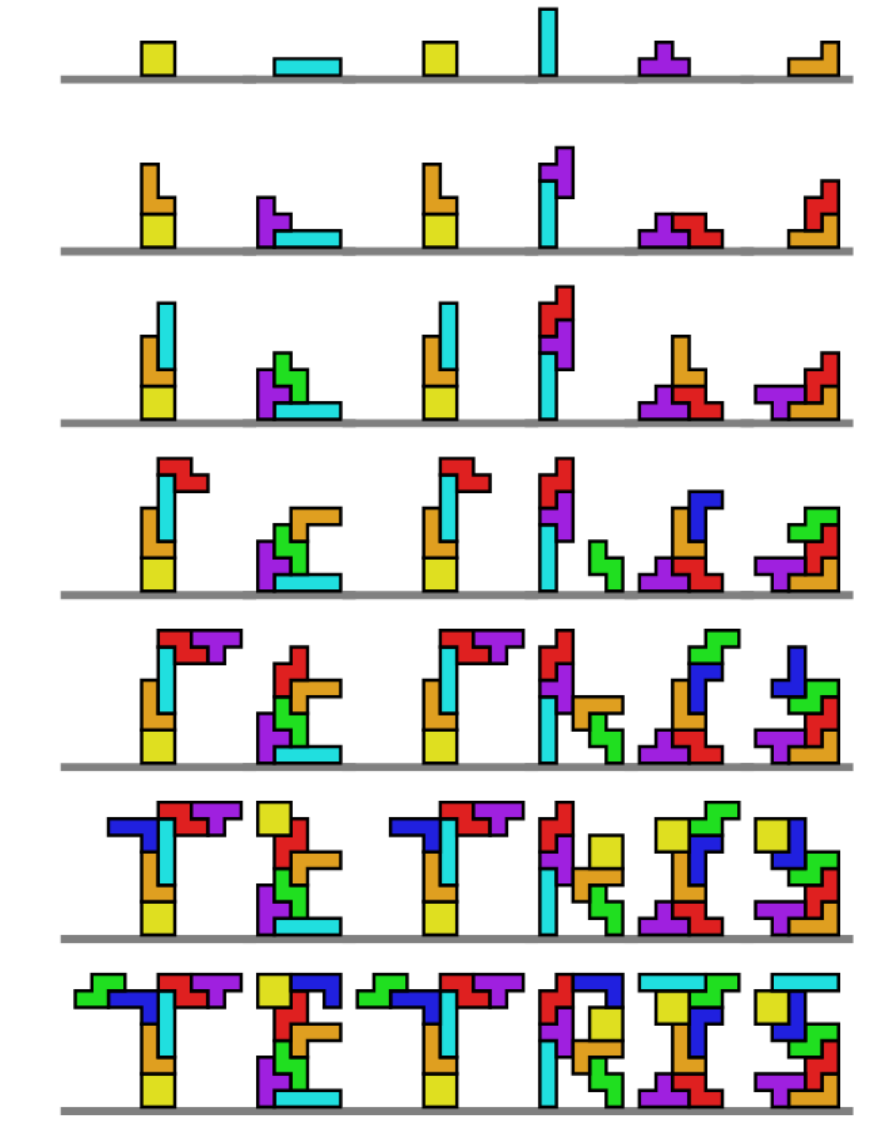
Richard Kaye
https://link.springer.com/article/10.1007/BF03025367

Demaine, Hohenberger, Liben-Nowell
https://erikdemaine.org/papers/Tetris_COCOON2003/paper.pdf

Tetris is NP-Hard

Tetris is NP-hard even with $O(1)$ rows or columns

Sualeh Asif* Michael Coulombe* Erik D. Demaine* Martin L. Demaine*
Adam Hesterberg* Jayson Lynch* Mihir Singhal*



29 Sep 2020
https://arxiv.org/pdf/2009.14336

Bejeweled, Candy Crush (…) are NP-Hard

Bejeweled, Candy Crush and other Match-Three Games are (NP-)Hard

L. Gualà[1], S. Leucci[2], and E. Natale[3]

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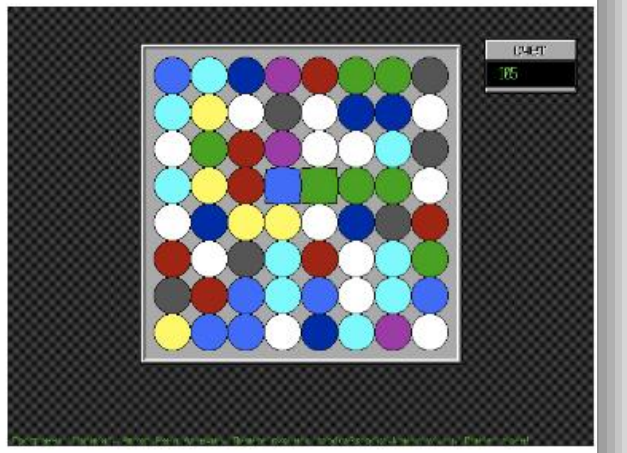
guala@mat.uniroma2.it
[2]Università degli Studi dell'Aquila
stefano.leucci@univaq.it
[3]*Sapienza* Università di Roma
natale@di.uniroma1.it

March 25, 2014

https://arxiv.org/pdf/1403.5830

(a) Candy Crush Saga

(b) Bejeweled



(c) Shariki

How to prove F is NP-Complete?

We have:

(Theorem 15.6 in the TCS Book):

For every $F \in \mathbf{NP}$, $F \leq_p$ **3SAT.**

Definition: A Boolean function G is **NP-Hard** if every $F \in \mathbf{NP}$ can be reduced to G: $F \leq_p G$.

Definition: A function G is **NP-Complete** if $G \in \mathbf{NP}$ and G is **NP-Hard**.

A function $F: \{0,1\}^* \to \{0,1\}$ is in **NP** if there exists some $a \in \mathbb{N}^+$ and $V: \{0,1\}^* \to \{0,1\}$ such that $V \in \mathbf{P}$ and $\forall x \in \{0,1\}^n$,
$F(x) = 1 \Leftrightarrow \exists w \in \{0,1\}^{n^a}$ such that $V(x,w) = 1$.

We want to show:

- F is in NP:

A function $F: \{0,1\}^* \to \{0,1\}$ is in **NP** if
there exists some $a \in \mathbb{N}^+$and $V: \{0,1\}^* \to \{0,1\}$
such that $V \in \mathbf{P}$ and $\forall x \in \{0,1\}^n$,
$F(x) = 1 \Leftrightarrow \exists w \in \{0,1\}^{n^a}$ such that $V(x,w) = 1$.

- F is NP-Hard:

Definition: A Boolean function G is **NP-Hard**
if every $F \in \mathbf{NP}$ can be reduced to G: $F \leq_p G$.

Equivalently:

Definition: A function G is **NP-Hard** if **3SAT** $\leq_p G$.

Cook-Levin Theorem (Theorem 15.6 in the TCS Book):

For every $F \in \mathbf{NP}$, $F \leq_p$ **3SAT.**

(Maximum) Independent Set

Definition: For any undirected graph $G = (V, E)$, we say a set $S \subseteq V$ is an *independent set* if for all $u, v \in S$, the pair $(u, v) \notin E$ is not an edge.

For any undirected graph G and integer k, the Boolean function $ISET$ is defined to be $ISET$(G,k) = 1 iff G has an independent set of size at least k.

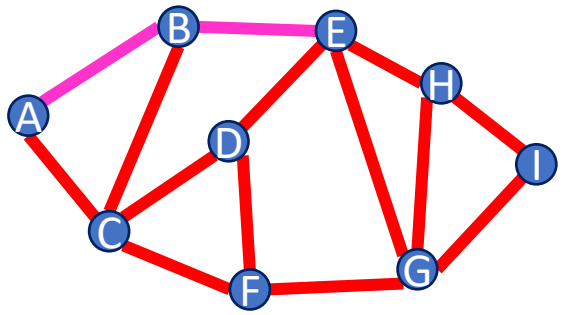
Example:

S1 = {A, I }

S2 = {A, E, F, I }

S3 = {B, D, G, H }

$ISET(G, 3)$ = $ISET(G, 6)$ =



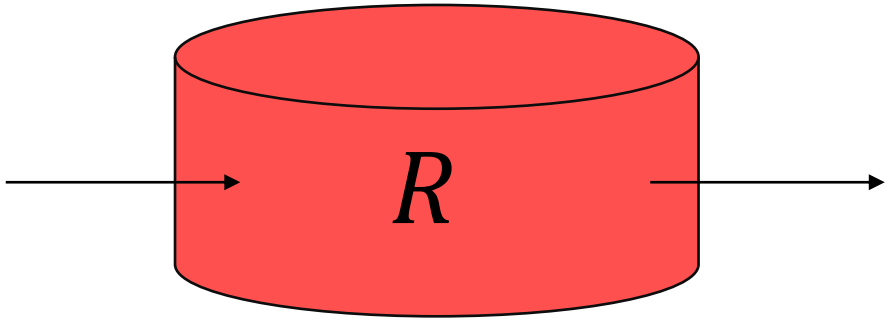
Is ISET NP-Complete?

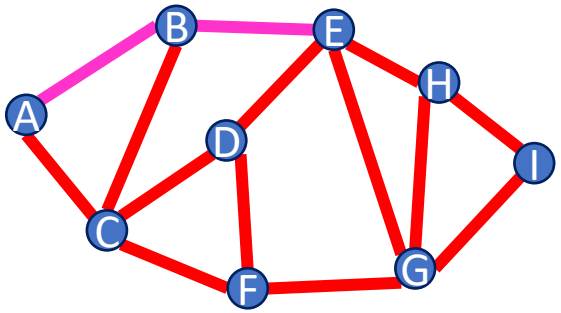
Max Independent Set:

For any undirected graph G and integer k, the Boolean function $ISET$ is defined to be $ISET$(G,k) = 1 iff G has an independent set of size at least k.

- Is $ISET$ in NP?
- Is $ISET$ NP-Hard? **3SAT** $\leq_p ISET$?
  What's the reduction?
  $R \in \mathbf{P}$ such that for all x, $3SAT(x) = ISET(R(x))$.

$(x_1 \vee x_2 \vee \bar{x}_3) \wedge (x_3 \vee \bar{x}_2 \vee x_4)$
$\wedge (\bar{x}_1 \vee x_2 \vee x_3)$



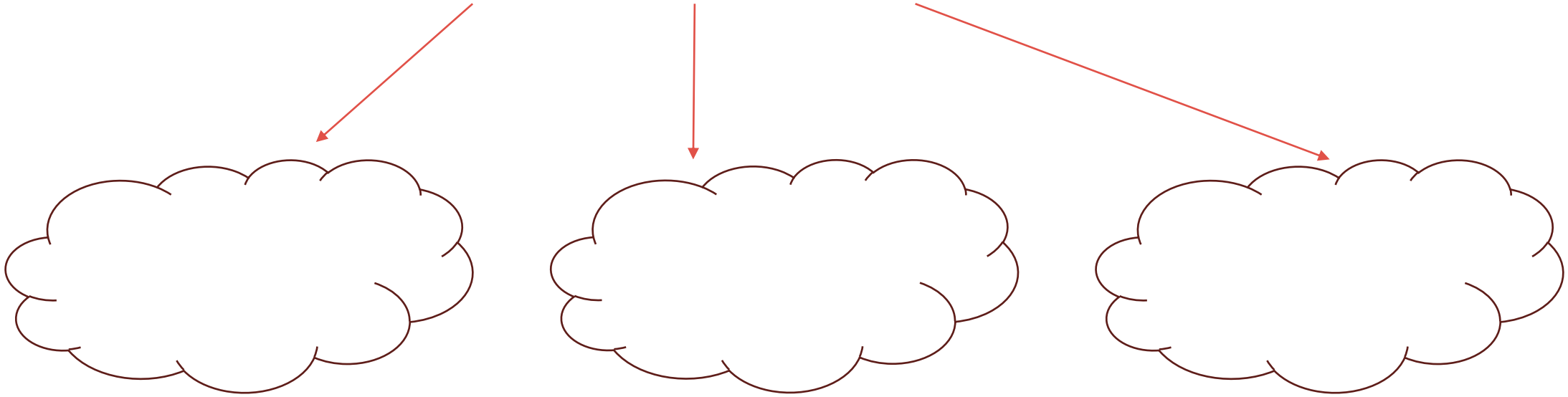


,k

How to reduce 3SAT to ISET?

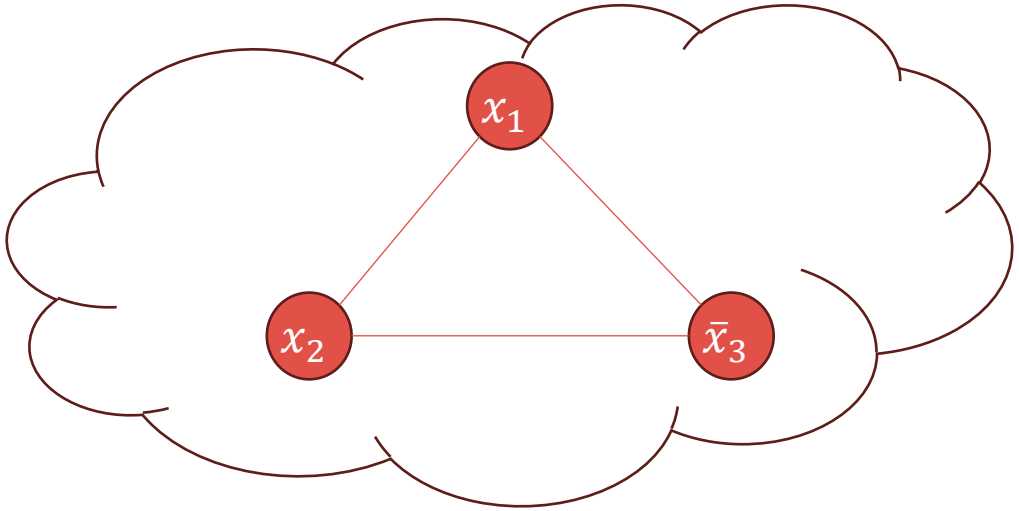
Idea: to map each clause into a “gadget” satisfying some (desired) properties.

$$(x_1 \vee x_2 \vee \bar{x}_3) \wedge (x_3 \vee \bar{x}_2 \vee x_4) \wedge (\bar{x}_1 \vee x_2 \vee x_3)$$



Gadget is triangle

$(x_1 \vee x_2 \vee \bar{x}_3)$



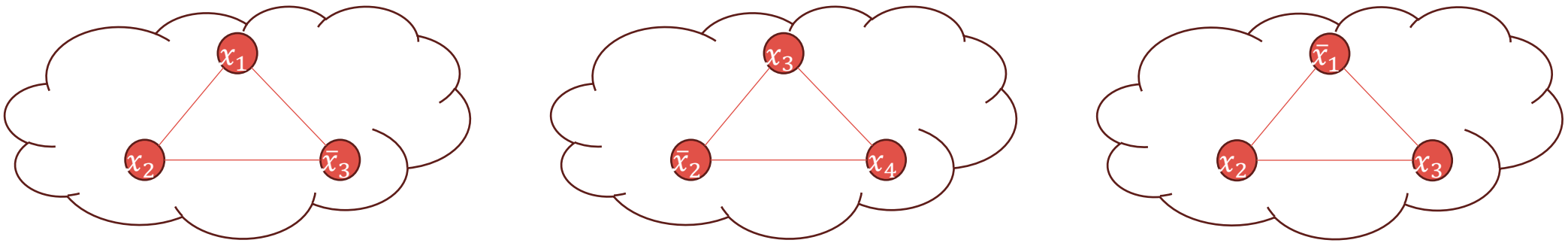
k = 1

Connect Gadgets

"Two literals in two clauses are negation of each other" is the global constraint

⇒ Put an edge as a constraint in ISET

$$(x_1 \vee x_2 \vee \bar{x}_3) \wedge (x_3 \vee \bar{x}_2 \vee x_4) \wedge (\bar{x}_1 \vee x_2 \vee x_3)$$



k = 3

Formalize the proof

Let x be a 3CNF, $x = c_1 \wedge c_2 \ldots \wedge c_n$ for some n.

For each $i \in [n]$, $c_i = a_{i1} \vee a_{i2} \vee a_{i3}$ for some literals a_{ij}.

Reduction $R(x)$ is constructed as follows.

$R(x)$:

1. Let $G = (V, E)$ be a graph of $3n$ vertices, $V = \{i1, i2, i3 : i \in [n]\}$.
2. The edges E is constructed as:
   a. For all $i \in [n]$, add edges $(i1, i2), (i2, i3), (i1, i3)$ to E
   b. For all $i, j \in [n]$, if $i \neq j$ and $a_{is} = \bar{a}_{jt}$ for some $s, t \in \{1,2,3\}$, add edge (is, jt) to E
3. Output (G, n)

Complete the proof with analysis

Let x be a 3CNF, $x = c_1 \wedge c_2 \ldots \wedge c_n$ for some n.

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3. Output (G, n)

Analysis:
If x is satisfiable, 3SAT(x) = 1

$\Leftrightarrow$ Exists assignment such that for each i, a_{is} = 1 for some s.

$\Leftrightarrow$ $U = \{is : a_{is} = 1, i \in [n]\}$ is an indep set of size n for R(x) bcs for all j, t s.t. $a_{is} = \bar{a}_{jt}$, jt is not in U.

$\Leftrightarrow$ ISET($R(x)$) = 1.

Hence, 3SAT(x) = (ISET($R(x)$) for all x. Also, we have $R \in \mathbf{P}$. Thus, 3SAT$\leq_p$ISET.

Framework of proofs

Precondition / notation

Let x be a 3CNF, $x = c_1 \wedge c_2 \ldots \wedge c_n$ for some n.

For each $i \in [n]$, $c_i = a_{i1} \vee a_{i2} \vee a_{i3}$ for some literals a_{ij}.

Reduction $R(x)$ is constructed as follows.

Algorithm of reduction

$R(x)$:

1. Let $G = (V, E)$ be a graph of $3n$ vertices, $V = \{i1, i2, i3 : i \in [n]\}$.
2. The edges E is constructed as:
   a. For all $i \in [n]$, add edges $(i1, i2), (i2, i3), (i1, i3)$ to E
   b. For all $i, j \in [n]$, if $i \neq j$ and $a_{is} = \bar{a}_{jt}$ for some $s, t \in \{1,2,3\}$, add edge (is, jt) to E
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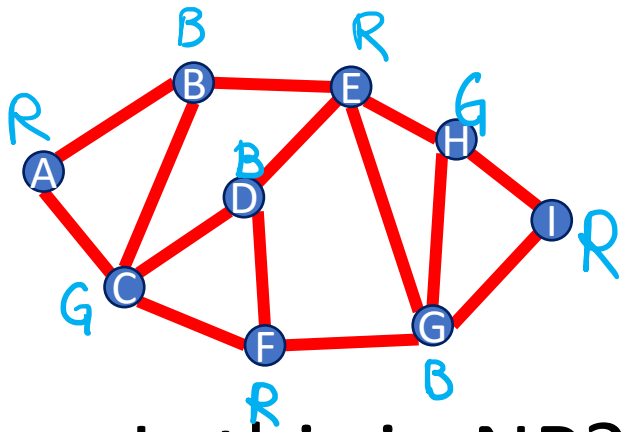
Hence, 3SAT(x) = (ISET($R(x)$) for all x. Also, we have $R \in \mathbf{P}$. Thus, 3SAT$\leq_p$ISET.

Analysis of reduction

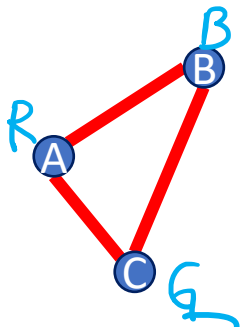
Graph 3-Coloring Problem

Can we color the vertices using 3 colors, each edge ends in exactly 2 different colors?

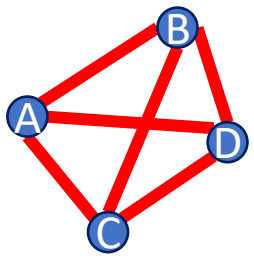
Yes



Yes



No



Is this in NP? Yes

$\exists w, V(G, w) = 1$ iff 3Color(G) $= 1$

coloring

$c(v) = \{R, B, G\}$

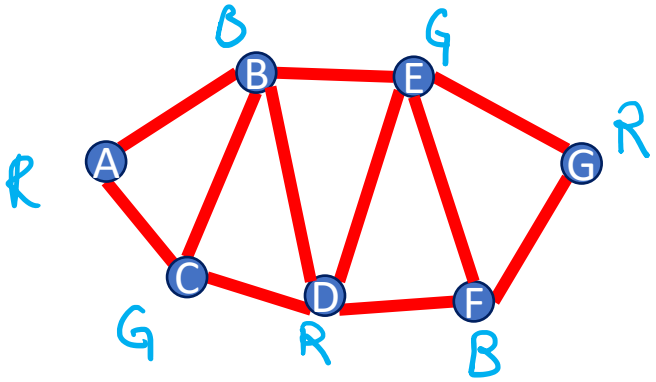
Is this NP-Complete?

3SAT $\leq_p$ 3Color

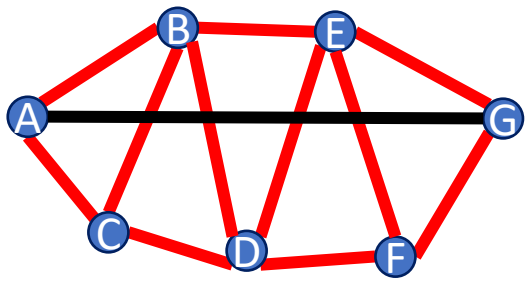
Exists 4-clique = Not 3-Colorable.

No 4-clique = 3-Colorable?

3-Colorable?



3-Colorable?

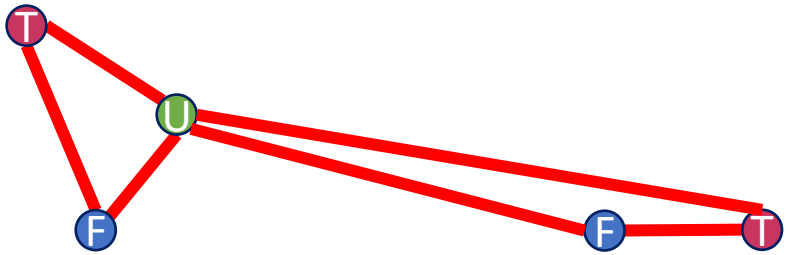


Reduction from 3SAT to 3-Coloring

For any 3CNF formula f, reduction R(f) outputs graph G s.t. 3SAT(f) = 3Coloring(R(f))

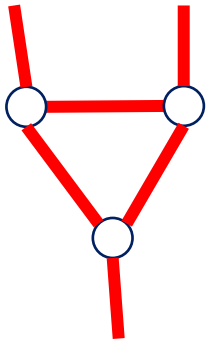
Gadgets:

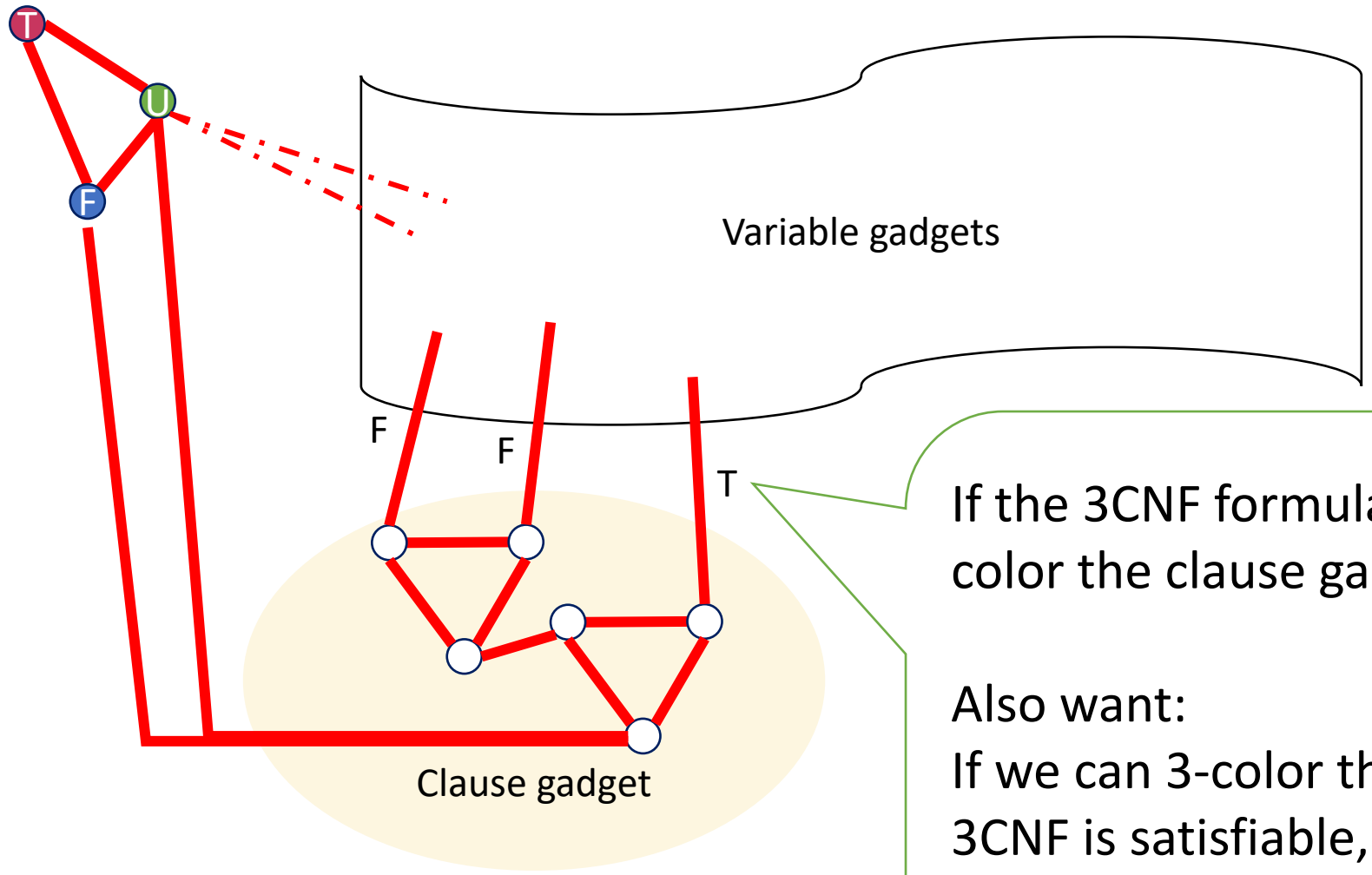
Constant



Variable

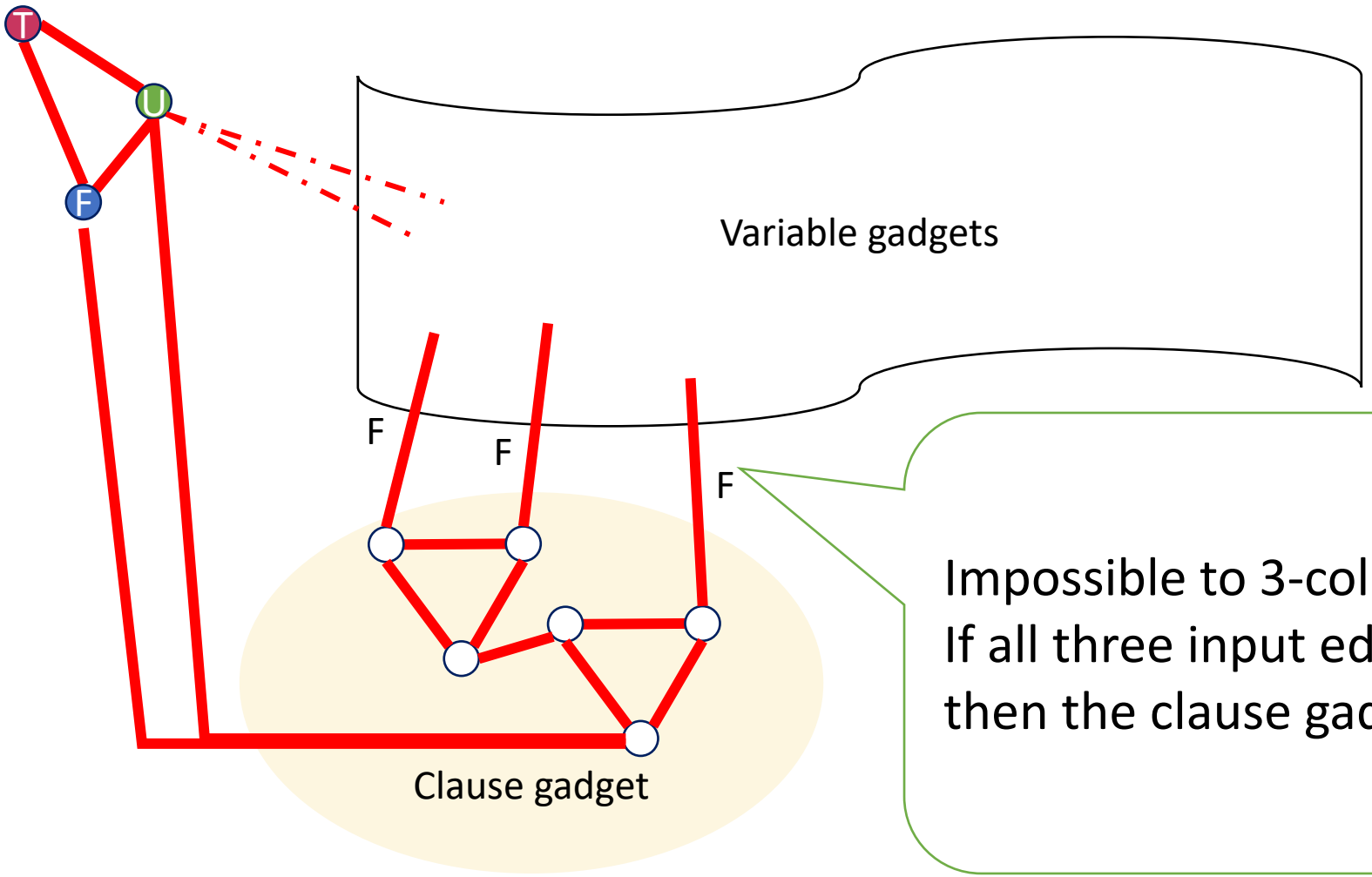
OR



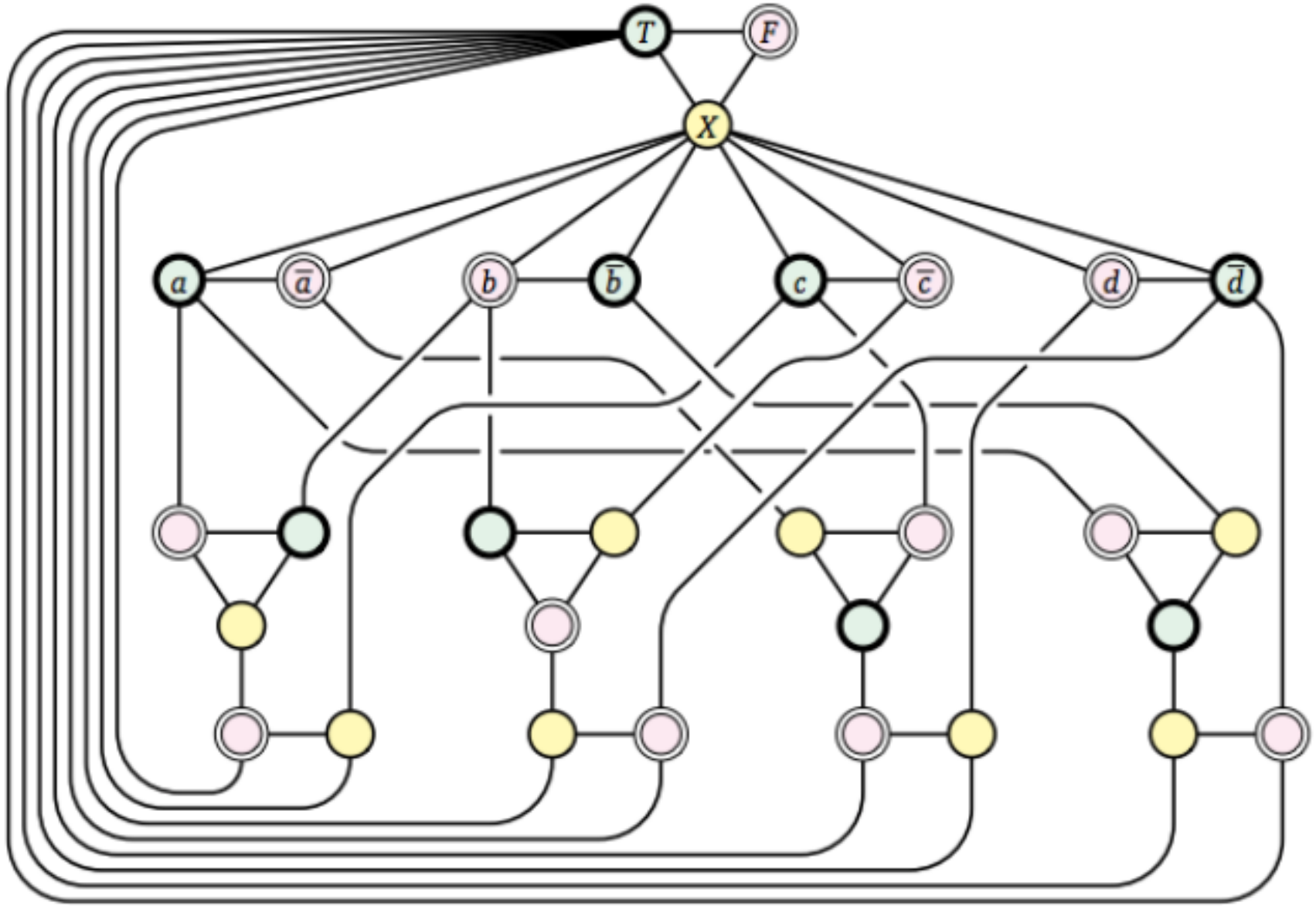


If the 3CNF formula is satisfiable, we can 3-color the clause gadget.

Also want:
If we can 3-color the clause gadget, then the 3CNF is satisfiable, ie, at least one input T.
See why on next page.



Impossible to 3-color:
If all three input edges are all F,
then the clause gadget is not 3-colorable.)



$$(a \vee b \vee c) \wedge (b \vee \bar{c} \vee \bar{d}) \wedge (\bar{a} \vee c \vee d) \wedge (a \vee \bar{b} \vee \bar{d})$$

https://www.cs.williams.edu/~shikha/teaching/fall19/cs256/lectures/Lecture24.pdf
See also:
https://cs.bme.hu/thalg/3sat-to-3col.pdf

Logistics

- **Final Exam: Monday, 4 May, 2:00pm - 4:00pm.** (We use only 2 hours.)

  120 min

- Previous: 20 min × 4 Mid ⇒ 40 min / section module
- 40 min NP, 20 min Card, reg, circuit, TM (total: 120)

HW 9 due next Monday.
Quiz 12 due next Monday.

Plan

Review for Final Exam

Gödel's Incompleteness Theorem

Textbook [TCS] Section 11.2

https://introtcs.org/public/lec_09_godel.html#g%C3%B6dels-incompleteness-theorem-computational-variant